

MEDIUM VOLTAGE · CAST RESIN · INDOOR

Instrument Transformers

Current & voltage transformers for metering and protection — LZZBJ9 & JDZX series
12 kV & 24 kV · IEC 61869



0.2S

METERING ACCURACY
CLASS

78.75 kA

DYNAMIC WITHSTAND ·
CT

100 VA

DUAL VT SECONDARIES

IEC

61869 SERIES
COMPLIANT

ENGINEERING THE FUTURE OF POWER WITH PRECISION AND INNOVATION

Since our inception in 2012, Powerline has been driven by a singular mission: to serve as the reliable backbone of the regional and Egyptian power sector. Starting as a trading company, Powerline has evolved into a trusted manufacturer of its own high-quality solutions — an ABB-certified panel builder for low voltage boards and an Original Equipment Manufacturer for medium voltage switchgear — backed by two factories and more than 2,200 delivered projects.

A Legacy of Reliability

Built on foundations of commitment and precision, Powerline has established itself as a hallmark of quality in major national and industrial projects. Our track record reflects a consistent ability to overcome complex engineering challenges and deliver customised solutions for the modern electrical grid.

Solutions Beyond Expectations

Our MV switchgear, Ring Main Units, compact substations and LV distribution boards are characterised by exceptional operational efficiency and a long life cycle — and our instrument transformers carry the same engineering DNA into measurement and protection.

Quality is not a department — it is our work culture



Production & Manufacturing

Advanced production lines ensure the highest accuracy in the assembly and calibration of MV & LV switchgear and components, in full compliance with IEC international standards.



Engineering & Design

Our design engineers develop layouts that prioritise maximum safety, operational continuity and seamless serviceability across the installation's life.



Testing & Quality Control

Every unit leaving our facilities undergoes rigorous routine testing to ensure durability under the harshest operating conditions — total peace of mind for our clients' investments.

2012

ESTABLISHED · CAIRO

2MANUFACTURING
FACILITIES**2,200+**

PROJECTS DELIVERED

OEMMV SWITCHGEAR · ABB-
CERTIFIED LV

“ Powerline: Performance Excellence, Execution Precision & Sustainable Solutions. ”

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How to use this catalogue

Sections 05–07 cover the current transformer range; sections 08–10 the voltage transformers. Section 11 lists routine tests and the information we need with your order.



CAST-RESIN INSTRUMENT TRANSFORMERS — VACUUM-ENCAPSULATED, FULLY ENCLOSED, MAINTENANCE-FREE

RANGE AT A GLANCE

Powerline’s indoor instrument transformers deliver accurate current and voltage measurement, energy metering and relay protection for medium voltage switchgear.

The primary and secondary windings, together with the core, are fully encapsulated under vacuum in cast epoxy resin — producing a compact, fully enclosed and maintenance-free unit with excellent insulation, low partial discharge, high moisture resistance and a long service life. The range covers the 12 kV (11 kV) and 24 kV (22 kV) classes for indoor metal-enclosed panels such as ring main units and MV switchgear, built to the IEC 61869 series.

Product	Type code	Class	Standard
Indoor current transformer	P-LZZBJ9-10	12 kV	IEC 61869-2
Indoor current transformer	P-LZZBJ9-24	24 kV	IEC 61869-2
Indoor voltage transformer	P-JDZX10-10	12 kV	IEC 61869-3
Indoor voltage transformer	P-JDZX11-20	24 kV	IEC 61869-3



LZZBJ9 current transformer · JDZX voltage transformer

Reference standards

IEC 61869-1 general requirements · IEC 61869-2 current transformers · IEC 61869-3 inductive voltage transformers.

12 / 24 kV

VOLTAGE CLASSES

2

CORES PER CT ·
WINDINGS PER VT

50 Hz

RATED FREQUENCY

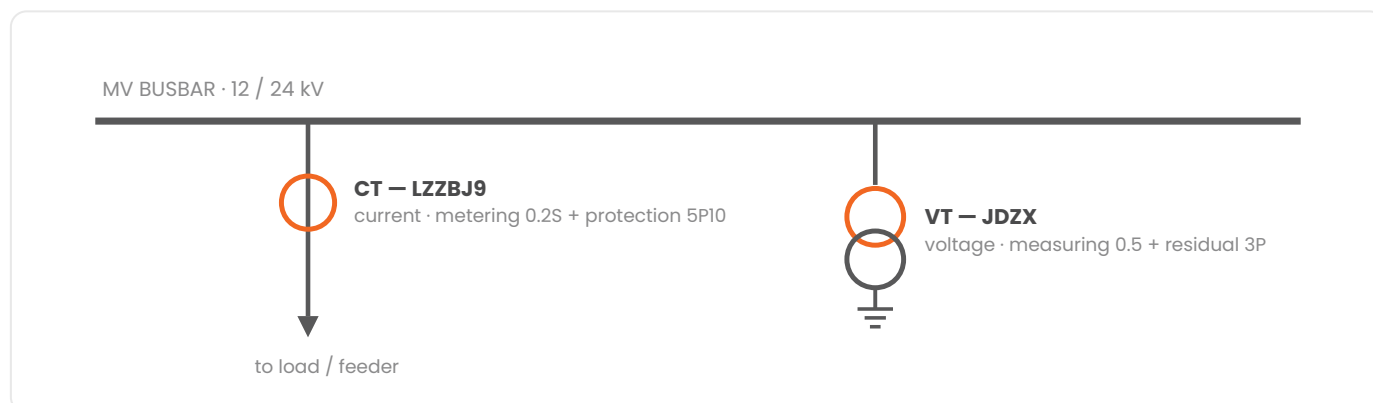
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MAINTENANCE
REQUIRED

Service conditions: indoor, metal-enclosed MV switchgear · 50 Hz · altitude ≤ 1,000 m (higher on request) · ambient -5...+40 °C · VTs for isolated / non-effectively earthed neutral systems.

FIELD OF APPLICATION

Wherever medium voltage energy must be measured, billed or protected, an instrument transformer translates the primary quantities to safe, standardised secondary signals — 5 A and 110 V — for meters, instruments and protection relays.



CT in series with the feeder · VT phase-to-earth on the busbar — metering and protection from one panel

Typical duties

- Revenue and check metering — class 0.2S metering cores
- Relay protection — 5P10 protection cores and 3P residual windings
- Earth-fault detection in isolated-neutral networks via open-delta (da-dn) windings
- Panel instrumentation: ammeters, voltmeters, energy meters and transducers

Typical environments

- MV metal-enclosed switchgear
- Ring main units
- Metering cubicles (PMC)
- Compact secondary substations
- Utility metering points
- Industrial substations

One family, one supplier

These are the same instrument transformers factory-fitted in the metering cubicles of Powerline's PSEC and PRAL ring main units and PCSS compact substations — engineered, installed and tested by one manufacturer.

CAST-RESIN TECHNOLOGY

Windings and core are cast in epoxy resin under vacuum — one solid, void-free insulation body that defines the performance of the whole range.



Full encapsulation

Primary and secondary windings and the core are completely enclosed in epoxy — a compact, robust, fully insulated block.



Low partial discharge

Vacuum casting eliminates voids in the insulation, keeping partial discharge low and dielectric life long.



Moisture resistant

The sealed epoxy body is unaffected by humidity and condensation — suited to harsh indoor service conditions.



Maintenance-free

No oil, no gas, nothing to inspect or refill — install it and forget it for the service life of the panel.



Compact mounting

Post-type CTs base-mounted on two bolts; block-type single-pole VTs — engineered for centre-mounted MV panels.



Long service life

Stable epoxy insulation, low losses and high mechanical strength deliver decades of accurate service.

Safety & handling

Current transformers

Never open-circuit the secondary while the primary is energised — a dangerous high voltage will appear across the terminals. Short unused secondary cores at their terminals.

Voltage transformers

Never short-circuit the secondary windings. Protect primary and secondary circuits with appropriately rated fuses — the terminal box is fuse-ready.

LZZBJ9 CURRENT TRANSFORMERS

Epoxy-resin cast, fully enclosed post-type current transformers for indoor switchgear — two independent cores and a reconnectable dual-ratio primary in one compact body.

Type designation

P

L

ZZ

B

J9

-10 / 24

P	Powerline product	B	Fully-enclosed construction
L	Current transformer	J9	Reinforced design, series 9
ZZ	Epoxy-resin cast, post (pillar) type	10/24	Rated voltage class (kV)

Key features

- Dual core:** separate metering (0.2S · 10 VA) and protection (5P10 · 15 VA) cores in one body
- Reconnectable ratio:** 600 / 1200 A primary selected via secondary links
- High short-circuit strength:** up to 78.75 kA dynamic, 31.5 kA · 3 s thermal
- Post mounting:** base-mounted on two bolts for centre-mounted MV panels
- Designed and tested to IEC 61869-2



P-LZZBJ9 enclosure designs — identical ratings

600-1200

/ 5 A DUAL RATIO

0.2S

METERING CLASS

5P10

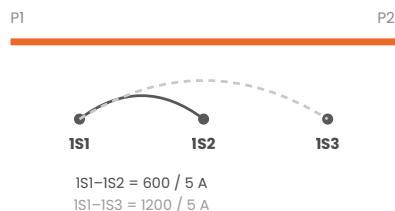
PROTECTION CLASS

FS ≤ 10

INSTRUMENT SECURITY

CT TECHNICAL DATA

Parameter	P-LZZBJ9-10	P-LZZBJ9-24
Reference standard	IEC 61869-2	
Installation · frequency	Indoor · 50 Hz	
Rated voltage class	12 kV (11 kV)	24 kV (22 kV)
Rated insulation level Um / AC / BIL	12 / 42 / 75 kV	24 / 65 / 125 kV
Rated current ratio	600 – 1200 / 5 A	600 – 1200 / 5 A
Core 1 — metering	0.2S · 10 VA	0.2S · 10 VA
Core 2 — protection	5P10 · 15 VA	5P10 · 15 VA
Instrument security factor Fs	≤ 10	
Rated power factor cos φ	0.8	
Short-time thermal current I _{th}	31.5 kA / 3 s	25 kA / 3 s
Dynamic peak current I _{dyn}	78.75 kA	62.5 kA
Secondary taps	1S1–1S2 = 600/5 · 1S1–1S3 = 1200/5	



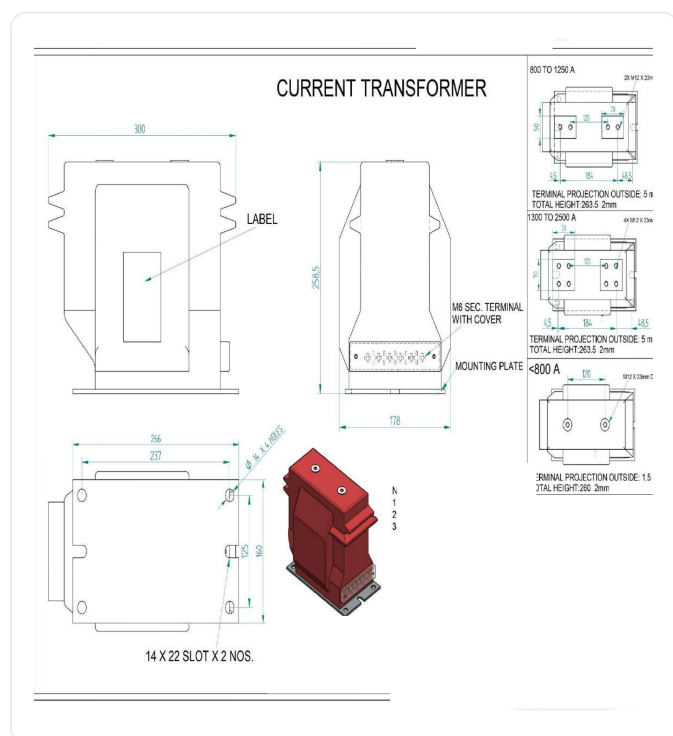
Reconnectable primary ratio via secondary links

One CT, two duties

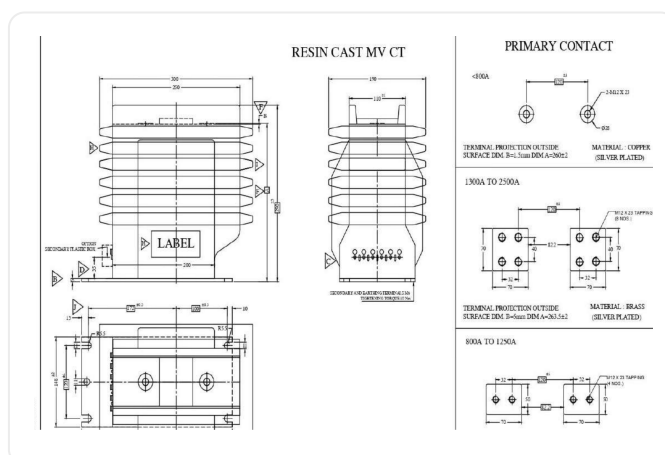
The 0.2S metering core keeps revenue metering accurate across the load range, while the 5P10 core feeds protection relays faithfully up to 10× rated current — no separate CTs, no extra panel space.

On request: special ratios, burdens, accuracy classes and additional cores are available per order — contact Powerline engineering.

CT DIMENSIONS & TERMINALS



P-LZZBJ9-10 — outline & mounting dimensions (mm)



P-LZZBJ9-24 — outline & primary contact details (mm)

Primary terminal	P-LZZBJ9-10	P-LZZBJ9-24
< 800 A	1 × M12 · H 260 mm	2 × M12 · H 260 mm
800 – 1250 A	2 × M12 · H 263.5 mm	4 × M12 · H 263.5 mm
1300 – 2500 A	4 × M12 · H 263.5 mm	8 × M12 · H 263.5 mm

Creepage: > 240 mm (12 kV) / > 440 mm (24 kV) ·

Torque: primary M12 — 50 N·m, secondary M6 — 10 N·m
· secondary terminals M6 with cover.

All dimensions in mm, third-angle projection. Dimensions are typical design values for continuous product improvement and are subject to confirmation per order.

JDZX VOLTAGE TRANSFORMERS

Single-phase, epoxy-resin cast, fully enclosed voltage transformers for indoor switchgear in isolated / non-effectively earthed neutral systems — measurement and earth-fault detection from one block.

Type designation

P

J

D

Z

X

10/11 · -10/-20

P	Powerline product	X	With residual (open-delta) winding
J	Voltage transformer	10/11	Design series
D	Single-phase	10/20	Rated voltage class (kV)
Z	Epoxy-resin cast		

Key features

- **Dual secondary:** measuring winding a–n plus residual winding da–dn for earth-fault detection
- **Two accuracy classes:** class 0.5 metering and class 3P protection — 100 VA each
- Designed for MV networks with isolated / non-effectively earthed neutral
- **Fuse-ready:** HV/LV terminal arrangement for primary and secondary protection
- Designed and tested to IEC 61869-3



P-JDZX enclosure designs — identical ratings

110/√3

MEASURING
SECONDARY · V

110/3

RESIDUAL
SECONDARY · V

0.5

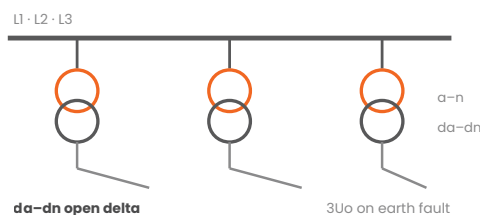
METERING CLASS

3P

PROTECTION CLASS

VT TECHNICAL DATA

Parameter	P-JDZX10-10	P-JDZX11-20
Reference standard	IEC 61869-3	
Type · frequency	Indoor, single-phase · 50 Hz	
Rated voltage class	12 kV (11 kV)	24 kV (22 kV)
Rated insulation level Um / AC / BIL	12 / 42 / 75 kV	24 / 65 / 125 kV
Rated primary voltage	11000 / $\sqrt{3}$ V	22000 / $\sqrt{3}$ V
Secondary voltage (a-n)	110 / $\sqrt{3}$ V	
Residual voltage (da-dn)	110 / 3 V	
Accuracy — measuring / protection	0.5 / 3P	
Rated output — measuring / residual	100 / 100 VA	
Rated power factor cos φ	0.8	
System earthing	Isolated / non-effectively earthed neutral	



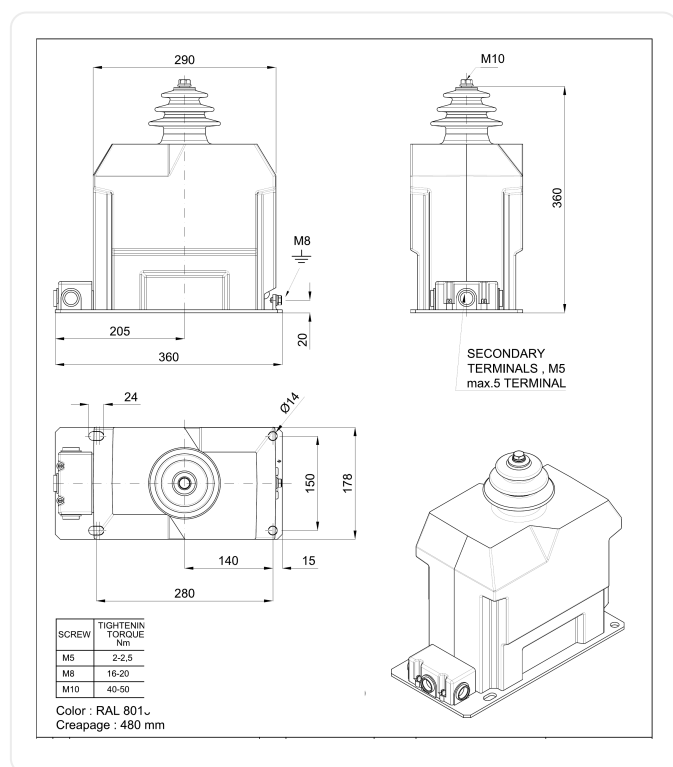
Three single-pole VTs — residual windings in open delta for earth-fault detection

Earth-fault detection built in

On a healthy network the open-delta voltage is near zero; an earth fault displaces the neutral and $3U_0$ appears across da-dn — driving the earth-fault relay without extra transformers.

On request: other primary voltages, burdens and accuracy combinations per order — contact Powerline engineering.

VT DIMENSIONS



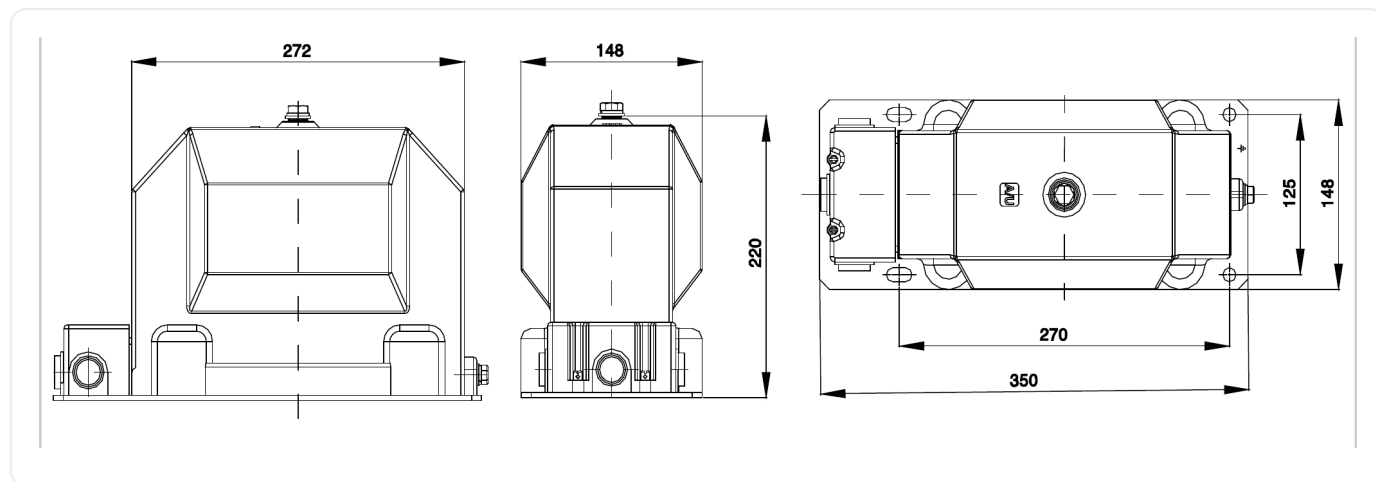
P-JDZX11-20 — outline & mounting dimensions (mm)

Screw	Tightening torque
M6 (secondary terminals, max. 5)	2 – 2.5 N·m
M8	16 – 20 N·m
M10 (primary)	40 – 50 N·m

Finish: RAL 8017 cast-resin body · **Creepage:** 480 mm (24 kV class) · single-pole block type, base-mounted.



P-JDZX — HV/LV terminal box with fuse provision



P-JDZX10-10 — outline dimensions (mm)

All dimensions in mm. Dimensions are typical design values for continuous product improvement and are subject to confirmation per order.

QUALITY, TESTING & ORDERING

Routine tests — every unit (IEC 61869)

- Verification of terminal markings and polarity
- Power-frequency withstand tests on primary and secondary windings
- Partial discharge measurement on the cast insulation
- Inter-turn overvoltage test
- Accuracy verification against the rated class — metering and protection cores / windings
- Documented test certificate delivered with every unit

Fitted and tested in our own panels

The same transformers are installed and secondary-tested in Powerline PSEC, PRAL and PCSS metering cubicles — ratios and burdens matched to the utility metering code at order stage.

Information required when ordering

#	Please specify
1	Product type and rated voltage class (kV)
2	Rated current ratio and number of cores (CT)
3	Rated primary / secondary voltage and windings (VT)
4	Accuracy class and rated burden per core / winding
5	Short-circuit ratings I _{th} / I _{dyn} (CT)
6	Mounting / panel type and quantity

Note. Losses, dimensions and ratings are typical design values in line with continuous product improvement and are confirmed per order. Special ratios, burdens and accuracy classes are available on request.



EVERY UNIT SHIPS WITH A DOCUMENTED TEST CERTIFICATE — ACCURACY, WITHSTAND AND PARTIAL DISCHARGE VERIFIED



Performance excellence. Execution precision. Sustainable solutions.

From engineering and manufacture to testing, delivery and lifetime support — Powerline is your integrated partner for MV & LV power distribution in Egypt and the region.

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For technical enquiries or quotations, please get in touch with our team — we respond with engineered solutions, not just prices.

12 / 24

KV VOLTAGE CLASSES

0.2S · 5P10

CT ACCURACY CLASSES

0.5 · 3P

VT ACCURACY CLASSES

IEC 61869

DESIGN & TESTING